

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

2nd Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination April May 2019

PHTCH006 Advanced Biopharmaceutics & Pharmacokinetics

Date: 27-04-2019, Saturday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 75

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1 (A) Describe the effect of formulation and processing factors on the dissolution of a tablet dosage form. **05**

(B) Write a brief note on in vitro-in vivo correlation. **05**

Q.2 Answer the followings: (Any three) **15**

(A) Describe the use of artificial intelligence in pharmaceuticals.

(B) Write a brief note on theories of drug dissolution.

(C) Describe the competition between drugs for the protein binding sites and their significance.

(D) Describe the factors affecting in vitro drug dissolution process.

Q.3 Answer the followings: (Any three) **15**

(A) Plasma drug concentration-time curve/profile for controlled release tablet dosage form

(B) Use of computational fluid dynamics in pharmaceuticals

(C) Model independent method of dissolution profile comparison

(D) One compartment open model for IV bolus administration

SECTION-II

Q.4 Explain factors affecting the drug absorption from gastrointestinal tract. **10**

Q.5 Describe the compendial dissolution methods as per USP. **10**

OR

Q.5 Describe in detail, the computational modelling of drug disposition. **10**

Q.6 Answer the followings: (Any three) **15**

- (A) Describe the pharmacokinetic methods useful in quantitative evaluation of bioavailability.
- (B) Write a note on biopharmaceutical classification system.
- (C) Describe the use of robotics in pharmaceuticals.
- (D) Explain the pharmacokinetic parameters that describe the plasma drug concentration-time curve.